

C: > Users > Afroz > Desktop > EXP5 > sdd.l

```
1  %{
2  #include <stdio.h>
3  #include <stdlib.h>
4
5  #define MAX 100
6
7  int values[MAX];
8  char operators[MAX];
9  int vtop = -1;
10 int otop = -1;
11
12 void pushValue(int value)
13 {
14     values[++vtop] = value;
15 }
16
17 int popValue()
18 {
19     return values[vtop--];
20 }
21
22 void pushOperator(char op)
23 {
24     operators[++otop] = op;
25 }
26
27 char popOperator()
28 {
29     return operators[otop--];
30 }
31
32 int precedence(char op)
33 {
34     if (op == '+' || op == '-')
35         return 1;
36
37     if (op == '*' || op == '/')
38         return 2;
39
40     return 0;
41 }
42
43 void applyOperator()
44 {
45     int a, b, result;
46     char op;
```

```
47
48     b = popValue();
49     a = popValue();
50     op = popOperator();
51
52     switch (op)
53     {
54         case '+':
55             result = a + b;
56             break;
57
58         case '-':
59             result = a - b;
60             break;
61
62         case '*':
63             result = a * b;
64             break;
65
66         case '/':
67             if (b == 0)
68             {
69                 printf("Error: Division by zero\n");
70                 exit(1);
71             }
72             result = a / b;
73             break;
74
75         default:
76             printf("Invalid operator\n");
77             exit(1);
78     }
79
80     pushValue(result);
81 }
82
83 void processOperator(char op)
84 {
85     while (otop >= 0 &&
86            operators[otop] != '(' &&
87            precedence(operators[otop]) >= precedence(op))
88     {
89         applyOperator();
90     }
91 }
```

```

92     pushOperator(op);
93 }
94
95 void processRightParenthesis()
96 {
97     while (otop >= 0 && operators[otop] != '(')
98     {
99         applyOperator();
100     }
101
102     if (otop < 0)
103     {
104         printf("Error: Mismatched parentheses\n");
105         exit(1);
106     }
107
108     popOperator();
109 }
110
111 %}
112 %%
113
114
115 [0-9]+ {
116     pushValue(atoi(yytext));
117 }
118
119 [+] {
120     processOperator('+');
121 }
122
123 [-] {
124     processOperator('-');
125 }
126
127 [*] {
128     processOperator('*');
129 }
130
131 [/] {
132     processOperator('/');
133 }
134
135 [(] {
136     pushOperator('(');
137 }

```

```

181 int main()
182 {
183     printf("Enter an arithmetic expression:\n");
184     yylex();
185     return 0;
186 }
187

```

```

138
139 [)] {
140     processRightParenthesis();
141 }
142
143 [\t]+ ;
144
145 \n {
146     while (otop >= 0)
147     {
148         if (operators[otop] == '(')
149         {
150             printf("Error: Mismatched parentheses\n");
151             exit(1);
152         }
153
154         applyOperator();
155     }
156
157     if (vtop == 0)
158     {
159         printf("Result = %d\n", values[vtop]);
160     }
161     else
162     {
163         printf("Error: Invalid expression\n");
164     }
165
166     vtop = -1;
167     otop = -1;
168 }
169
170 . {
171     printf("Invalid character: %s\n", yytext);
172 }
173
174 %%
175
176 int yywrap()
177 {
178     return 1;
179 }
180

```

```
M /c/Users/Afroz/Desktop/EXP5

Afroz@DESKTOP-TDQFQAL UCRT64 ~
$ cd /c/Users/Afroz/Desktop

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop
$

mkdir EXP5

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop
$ cd EXP5

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP5
$ nano sdd.1

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP5
$ ls
sdd.1

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP5
$ flex sdd.1

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP5
Enter an arithmetic expression:
10+5*2
Result = 20
|
```